

Exercise Sheet 4

Linear Models: Least Squares, Logistic Regression and Calibrated Scores

Probabilistic Machine Learning

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4.A Least Squares Is a Likelihood

Exercise 4.A.1: Four Flats and a Villa

Model $y = w_0 + w_1x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ i.i.d.; x in m², y in k€.

x	40	60	80	100
y	150	200	280	330

1. Write down $\ell(w_0, w_1, \sigma^2)$ and show that for **every** σ^2 the maximum over (w_0, w_1) is the least-squares fit.
2. Compute $\hat{w}_1 = S_{xy}/S_{xx}$ and $\hat{w}_0 = \bar{y} - \hat{w}_1\bar{x}$, the residuals and the RSS.
3. Compute $\hat{\sigma}_{\text{ML}}^2 = \text{RSS}/n$ and the unbiased $\text{RSS}/(n - 2)$. By which factor does the ML version underestimate σ^2 on average?
4. Give the plug-in 95% band $\hat{y} \pm 1.96\hat{\sigma}_{\text{ML}}$ at 80 m² and at 600 m². Then use the half-width that counts the uncertainty about $\mathbf{w}$, $1.96\hat{\sigma}\sqrt{1 + \frac{1}{n} + \frac{(x-\bar{x})^2}{S_{xx}}}$. What do you conclude?

4.B Classification: Only $p(y | x)$

Exercise 4.B.1: Cross-Entropy and Its Gradient

$h_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$ with $\sigma(z) = 1/(1 + e^{-z})$ and $\ell(\mathbf{w}) = \sum_i [y_i \ln h_i + (1 - y_i) \ln(1 - h_i)]$.

1. Show $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.
2. Show $\frac{\partial \ell}{\partial \mathbf{w}} = \sum_i (y_i - h_i) \mathbf{x}_i$.
3. Six transactions, $\mathbf{x} = (1, \text{new device})$: new device — fraud, fraud, legitimate; known device — fraud, legitimate, legitimate. Compute the gradient at $\mathbf{w} = 0$.
4. Set both components of the gradient to zero: which scores does the MLE give to new and to known devices? Give $\hat{w}_0, \hat{w}_1$ and read $\hat{w}_1$ as an odds ratio.

Exercise 4.B.2: The Peak That Runs Away (Python)

```
import numpy as np
x = np.array([-2.0, -1.5, -1.1, -0.6, -0.3, 0.2, 0.6, 1.0, 1.4, 1.9])
y = np.array([0, 0, 0, 0, 0, 0, 1, 1, 1, 1])
sigma = lambda z: 1 / (1 + np.exp(-z))

def fit(tau2=None, steps=10_000, eta=0.05):
    w0 = w1 = 0.0
    for t in range(1, steps + 1):
        h = sigma(w0 + w1 * x)
        g0 = ___ # gradient for the intercept
        g1 = ___ # gradient for the slope
        if tau2 is not None:
            g1 -= ___ # log-prior N(0, tau2) on the slope
        w0, w1 = w0 + eta * g0, w1 + eta * g1
        if t in (10, 100, 1_000, 10_000):
            h = sigma(w0 + w1 * x)
            print(t, round(w1, 2), np.sum(y * np.log(h) + (1 - y) * np.log(1 - h)))
```

1. Fill in the blanks and run `fit()`. What happens to the slope, the log-likelihood and the scores? Why will it never stop?
2. Run `fit(tau2=1)` and `fit(tau2=100, steps=100_000)`. Explain the difference with the MAP formula of the lecture.

4.C Generative vs Discriminative

Exercise 4.C.1: The Same Line

Naive Bayes with yes/no flags, $\theta_{jy} = P(x_j = 1 \mid y)$, prior $\pi = P(y = 1)$. Lecture numbers: $\pi = 1/1000$; new device 0.6 vs 0.1; foreign IP 0.7 vs 0.05; night 0.5 vs 0.2 (fraud vs honest).

1. Show that $\log \frac{P(y=1|x)}{P(y=0|x)} = w_0 + \sum_j w_j x_j$ and give w_j and w_0 .
2. Compute the three weights and w_0 , and check that a transaction with all three flags gets $P(\text{fraud} \mid x) \approx 17.4\%$.
3. The foreign-IP flag is **missing**. Compare what naive Bayes should do with plugging $x_{\text{IP}} = 0$ into the linear form.
4. Why can logistic regression, which has the same linear form, not do what naive Bayes does in 3?

Exercise 4.C.2: Counting Twice, in Code

Complete the three marked lines (numpy only), then run the script.

```
import numpy as np
rng = np.random.default_rng(42)
n = 5000
y = (rng.random(n) < 0.3).astype(int)          # 1 = fraud
flag = lambda pf, ph: (rng.random(n) < np.where(y == 1, pf, ph)).astype(int)
device = flag(0.6, 0.1)
copy = np.where(rng.random(n) < 0.95, device, 1 - device)  # a near-copy
night = flag(0.5, 0.2)
X = np.column_stack([device, copy, night])

def fit_nb(X, y, alpha=1.0):
    prior = ...                # (a)  $P(y = 1)$ 
    th1 = ...                   # (b) Laplace-smoothed  $P(x_j = 1 \mid y = 1)$ 
    th0 = ...                   # and  $P(x_j = 1 \mid y = 0)$ 
    w = np.log(th1 / th0) - np.log((1 - th1) / (1 - th0))
    w0 = np.log(prior / (1 - prior)) + np.log((1 - th1) / (1 - th0)).sum()
    return w0, w

def fit_lr(X, y, steps=5000, lr=0.5):  # gradient ascent, as in the lecture
    Xb = np.column_stack([np.ones(len(X)), X]); w = np.zeros(Xb.shape[1])
    for i in range(steps):
```

1. Print w_0 , w and $P(\text{fraud} \mid x = (1, 1, 1))$ for both models.
2. Compare the weights of device and copy. Which model counts the device twice, and which shares the credit?
3. Which probability is closer to the fraction of frauds among the training rows with $x = (1, 1, 1)$?

4.D When Is a Score a Probability?

Exercise 4.D.1: The Rebalanced Fraud Model

A model is trained on a 50/50 sample; on real traffic fraud is **1 in 1,000**. Undersampling kept every fraud and a random share of legitimate cases.

1. Use Bayes' theorem in odds form to show: $\text{real odds} = \text{model odds} \times \frac{1/999}{1}$. Which assumption about $p(\mathbf{x} \mid y)$ do you need?
2. Convert the scores 0.5, 0.9 and 0.99 into real fraud probabilities.
3. Blocking a legitimate card costs €10, missing a fraud €500. Above which real probability should a card be blocked, and which **raw score** is that?

Exercise 4.D.2: Weight Decay Is a Prior

1. With $y_i = \mathbf{w}^\top \mathbf{x}_i + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ and a prior $\mathbf{w} \sim \mathcal{N}(0, \tau^2 \mathbf{I})$, show that the MAP minimises $\text{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|^2$ with $\lambda = \sigma^2 / \tau^2$.
2. One weight, no intercept, $y = wx + \varepsilon$. Show $\hat{w}_{\text{MAP}} = \frac{\sum_i x_i y_i}{\sum_i x_i^2 + \lambda}$.
3. Data $(x, y) = (1, 2), (3, 7)$, $\sigma^2 = 1$, $\tau^2 = 0.1$. Compute the MLE and the MAP. What happens as $\tau^2 \rightarrow \infty$ and as $\tau^2 \rightarrow 0$?